1. Develop a Program in C for the following: a) Declare a calendar as an array of 7 elements (A dynamically Created array) to represent 7 days of a week. Each Element of the array is a structure having three fields. The first field is the name of the Day (A dynamically allocated String), The second field is the date of the Day (A integer), the third field is the description of the activity for a particular day (A dynamically allocated String).

b) Write functions create(), read() and display(); to create the calendar, to read the data from the keyboard and to print weeks activity details report on screen.

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

typedef struct {
   char *name;       // Name of the day
   int date;         // Date of the day
   char *description; // Description of the activity
} Day;

void create(Day *week, int size) {
   for (int i = 0; i < size; i++) {
      week[i].name = (char *)malloc(20 * sizeof(char)); // Allocate memory for name
      week[i].description = (char *)malloc(100 * sizeof(char)); // Allocate memory for description
   }
}

void read(Day *week, int size) {
   for (int i = 0; i < size; i++) {
      printf("Enter name of day %d: ", i + 1);
      scanf("%s", week[i].name);
      printf("Enter date: ");
      scanf("%d", &week[i].date);
      printf("Enter description: ");
      scanf(" %[^\n]", week[i].description); // Read a line of text including spaces
   }
}

void display(Day *week, int size) {
   printf("\nWeek's Activity Details:\n");
```

```
    for (int i = 0; i < size; i++) {
        printf("Day: %s, Date: %d, Activity: %s\n", week[i].name, week[i].date, week[i].description);
    }
}

int main() {
    int size = 7; // 7 days in a week
    Day *week = (Day *)malloc(size * sizeof(Day)); // Dynamically allocate the array

    create(week, size); // Allocate memory for names and descriptions
    read(week, size);   // Read data from user
    display(week, size); // Display the week's activities

    // Free the allocated memory
    for (int i = 0; i < size; i++) {
        free(week[i].name);
        free(week[i].description);
    }
    free(week);
    return 0;
}
```

```
Enter name of day 1: monday
Enter date: 21
Enter description: team meeting
Enter name of day 2: tuesday
Enter date: 22
Enter description: presentation
Enter name of day 3: wednesday
Enter date: 23
Enter description: work from home
Enter name of day 4: thursday
Enter date: 24
Enter description: project deadline
Enter name of day 5: friday
Enter date: 25
Enter description: team outing
Enter name of day 6: saturday
Enter date: 26
Enter description: submission report
Enter name of day 7: sunday
Enter date: 27
Enter description: rest day

Week's Activity Details:
Day: monday, Date: 21, Activity: team meeting
Day: tuesday, Date: 22, Activity: presentation
Day: wednesday, Date: 23, Activity: work from home
Day: thursday, Date: 24, Activity: project deadline
Day: friday, Date: 25, Activity: team outing
Day: saturday, Date: 26, Activity: submission report
Day: sunday, Date: 27, Activity: rest day
```

Input : Enter name of day 1: Monday
Enter date: 1
Enter description: Team meeting at 10 AM

Enter name of day 2: Tuesday
Enter date: 2
Enter description: presentation

Enter name of day 3: Wednesday
Enter date:
Enter description: Work from home

Enter name of day 4: Thursday
Enter date: 4
Enter description: Project deadline

Enter name of day 5: Friday
Enter date: 5
Enter description: Team outing

Enter name of day 6: Saturday
Enter date: 6
Enter description: submission report

Enter name of day 7: Sunday
Enter date: 7
Enter description: Rest day

Output:
Week's Activity Details:
Day: Monday, Date: 1, Activity: Team meeting at 10 AM
Day: Tuesday, Date: 2, Activity: Client presentation
Day: Wednesday, Date: 3, Activity: Work from home
Day: Thursday, Date: 4, Activity: Project deadline
Day: Friday, Date: 5, Activity: Team outing
Day: Saturday, Date: 6, Activity: Family time
Day: Sunday, Date: 7, Activity: Rest day